DO NOW — RECALL

Stacks and queues

Name: _____ Class: _____ Date: _____

1. What is a[1] for int[] a={6,9,3}?

2. What is −1 in 8-bit two's complement?

3. A loop running from 0 to 3 inclusive executes how many times?

4. Push 2, push 5, pop once: which value is removed?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

9.

2. Model answer

11111111.

3. Model answer

4.

4. Model answer

5, the last item pushed.

Vocabulary — know these terms

stack — LIFO abstract data type.

push — Add an item onto a stack.

pop — Remove the top stack item.

queue — FIFO abstract data type.

dequeue — Remove the front queue item.

NEW LEARNING

Read, annotate and explain

Push

Adding an item puts it on top of a stack.

Pop

Removing an item takes the most recently pushed item: 7.

Contrast with a queue

A queue removes the oldest item instead (FIFO); it is suited to arrival-order processing.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A stack receives 3, 8, 5, then Pop once. Give removed value and remaining stack bottom-to-top.

Predict

The newest value 5 is on top, so Pop removes 5.

Confirm and explain

The stack remains [3,8] from bottom to top.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Push A, B, C onto a stack then Pop twice. What remains?

Question 2

A circular array queue has capacity 3. Starting empty, enqueue 10,20,30; dequeue once; enqueue 40. State physical slots, front/rear/count and logical order. Use rear as the next free slot.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why an undo history usually uses a stack rather than a queue.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A queue removes the most recently inserted item first.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.